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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 08

B.Tech. (EE) PT (Sem.–1)

ENGINEERING MATH-III

Subject Code : BTAM-301

M.Code : 70970

Time : 2 Hrs. **Max. Marks : 30**

INSTRUCTIONS TO CANDIDATES :

1. **Attempt any FIVE question(s), each question carries 6 marks.**

1. Use the concept of residues to evaluate $\int_0^{2\pi} \frac{d\theta}{(5-3\cos\theta)^2}$.

2. Find a Fourier series to represent, $x - x^2$ from $x = -\pi$ to $x = \pi$. Hence, show that

$$\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+.....=\frac{\pi^2}{12}.$$

3. Find $L^{-1}\left[\log\left(1+\frac{1}{s^2}\right)\right]$, where L^{-1} denotes inverse Laplace transform.

4. If $f(z) = u + iv$ is an analytic function of $z = x + iy$ and $u - v = \frac{\cos x+\sin x-e^{-y}}{2\cos x-e^{y}-e^{-y}}$, find the analytic function $f(z)$, given $f\left(\frac{\pi}{2}\right)=0$.

5. Solve the equation by method of separation of variables.

$$4\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}=3u \text{ given } u=3e^{-y}-e^{-5y} \text{ when } x=0.$$

6. Using the recurrence relation $(n + 1)P_{n+1}(x) = (2n + 1)\, xP_n(x) - nP_{n-1}(x)$. Hence, evaluate $P_2(1.5)$ and $P_3(2.1)$.

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7. A tightly stretched string has its ends fixed at $x = 0$ and $x = l$. At time $t = 0$ the string is given a shape defined by $f(x) = \mu x\,(l - x)$, where μ is constant and then released. Find the displacement of any point x of the string at any time $t > 0$.

8. Find the power series solution about the origin of the equation :

$$\left(1 - x^2\right) y'' - 4xy' + 2y = 0.$$

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